

***“Analyzing Job Market Trends and Employer Sentiment
using NLP and BERT on Glassdoor Data Science Job
Postings”***

Subhankar Biswas

***MDSSB-MET-02_Text Analysis and Natural Language Processing
Project***

Constructor University, Bremen, Germany

Email: subiswas@constructor.university

Matriculation Number: 30008967

Abstract

The rapid growth of online recruitment platforms has generated a massive amount of textual job-related data that can be analysed using Natural Language Processing (NLP) and Machine Learning techniques. This project investigates Glassdoor job postings to analyse employer sentiment, job market trends, and semantic structures within job descriptions using both classical NLP methods and transformer-based deep learning models. The project combines preprocessing, TF-IDF feature extraction, topic modelling, clustering, sentiment classification, and BERT-based contextual learning.

The dataset consists of Glassdoor job postings containing textual job descriptions, salary estimates, company ratings, and other structured attributes. Data preprocessing techniques including lowercasing, tokenization, stopwords handling, and text normalization were applied to clean and standardize textual information. TF-IDF vectorization was used to transform job descriptions into numerical feature vectors suitable for machine learning models.

Latent Dirichlet Allocation (LDA) was implemented for topic modelling to identify latent semantic structures in job postings, while KMeans clustering grouped similar postings according to textual similarity. Sentiment labels were generated using company ratings as proxy labels, where ratings above or equal to 3.5 were considered positive and ratings below 3.5 were considered negative.

Two classical machine learning models, Logistic Regression and Naive Bayes, were implemented for sentiment analysis. Additionally, BERT (Bidirectional Encoder Representations from Transformers) was fine-tuned for transformer-based sentiment classification. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix analysis.

Experimental results demonstrated that transformer-based models significantly outperform classical NLP approaches due to superior contextual understanding capabilities. However, the project also highlights challenges such as noisy labels, class imbalance, and limitations of proxy sentiment labelling strategies.

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Keywords: Natural Language Processing, BERT, Sentiment Analysis, Topic Modelling, TF-IDF, Text Mining, Machine Learning, Glassdoor.

1. Introduction

The digitization of recruitment platforms has transformed the modern employment ecosystem. Websites such as Glassdoor, LinkedIn, Indeed, and Monster generate large volumes of textual information related to job postings, company reviews, employee feedback, and salary estimates. These textual datasets provide valuable insights into labour market trends, employer expectations, and organizational sentiment.

Natural Language Processing (NLP) has become an essential computational approach for analyzing such large-scale textual datasets [AB1999], [AB2016]. By combining linguistic processing techniques with machine learning algorithms, NLP enables automated extraction of semantic patterns, hidden topics, and sentiment indicators from unstructured textual information.

This project focuses on the analysis of Glassdoor job postings using NLP and transformer-based deep learning models. The study explores multiple objectives including:

- identifying hidden semantic topics in job descriptions,
- clustering similar job postings,
- analyzing employer sentiment,
- comparing classical NLP models with BERT transformers.

The project follows a complete text mining pipeline beginning with data preprocessing and extending to advanced transformer-based sentiment analysis. Classical machine learning models including Logistic Regression and Naive Bayes are compared against BERT-based contextual models.

The significance of this project lies in its interdisciplinary integration of:

- Natural Language Processing,
- Machine Learning,
- Deep Learning,
- Text Mining,
- Statistical Analysis.

Furthermore, the project demonstrates the practical application of NLP techniques in labour market analytics and recruitment intelligence.

2. Research Objectives

The primary objectives of this project are:

1. To preprocess and clean Glassdoor job posting text data.
2. To extract meaningful textual features using TF-IDF vectorization.
3. To identify hidden semantic structures using topic modelling.
4. To cluster similar job postings using KMeans clustering.
5. To perform sentiment analysis using classical machine learning techniques.
6. To fine-tune a BERT transformer model for contextual sentiment analysis.
7. To compare the performance of classical NLP models and transformer-based models.
8. To evaluate model effectiveness using multiple evaluation metrics.

3. Research Questions

The project addresses the following research questions:

1. Can NLP techniques effectively identify semantic trends within Glassdoor job postings?
2. How effectively can topic modelling identify hidden themes in recruitment data?
3. Can job postings be clustered into semantically meaningful groups?
4. How well do classical machine learning models perform on sentiment classification tasks?
5. Does BERT significantly outperform traditional NLP approaches for employer sentiment analysis?
6. What are the primary limitations of using company ratings as proxy sentiment labels?

4. Dataset Description

4.1 Dataset Source and Collection Method

One of the key considerations in this project was the selection of a reliable and reproducible dataset source. The project uses an existing publicly available Glassdoor Job Postings dataset obtained from Kaggle rather than performing independent web scraping. This decision was made for several important reasons.

First, publicly available datasets provide reproducibility and transparency, which are critical requirements for academic research. Since the dataset is already curated and structured, other researchers and evaluators can reproduce the experimental pipeline without legal or ethical concerns associated with live web scraping.

Second, direct web scraping from Glassdoor may violate platform policies and create challenges related to data consistency, incomplete scraping, duplicate records, and changing webpage structures. Using an established dataset ensures stable experimentation and standardized preprocessing.

The dataset contains a collection of Glassdoor job postings including job descriptions, salary estimates, company ratings, industries, and company-related metadata. The dataset was downloaded and stored locally inside the project repository under:

```
ext.{data/raw/Glassdoor_Job_Postings.csv}
```

The project therefore focuses on NLP analysis, machine learning experimentation, and transformer-based modelling rather than large-scale data acquisition.

This clarification is important because the project objective is centred on text mining and sentiment analysis methodology rather than data scraping infrastructure.

4. Dataset Description

The dataset used in this project consists of Glassdoor job postings collected from publicly available recruitment datasets. The dataset contains structured and unstructured information including:

Feature	Description
job_description	Textual description of job postings
company	Company name
location	Job location
salary_avg_estimate	Estimated salary
company_rating	Company rating
industry	Industry category
sector	Business sector

The dataset contains both textual and numerical attributes, making it suitable for combined NLP and machine learning analysis.

4.1 Sentiment Label Generation

Since explicit sentiment labels were not available, proxy sentiment labels were generated using company ratings.

The sentiment labelling strategy was defined as:

- Positive sentiment $\rightarrow$ company_rating ≥ 3.5
- Negative sentiment $\rightarrow$ company_rating < 3.5

Although this approach introduces noise and potential bias, it provides a practical approximation for sentiment classification tasks.

5. Data Preprocessing

Text preprocessing is a critical stage in NLP pipelines because raw textual data often contains noise, inconsistencies, punctuation, symbols, and irrelevant information.

The preprocessing pipeline implemented in this project includes:

5.1 Lowercasing

All textual data was converted to lowercase to ensure case consistency.

Example:

Original Text:

"Senior Data Scientist"

Processed Text:

"senior data scientist"

5.2 Special Character Removal

Regular expressions were used to remove:

- punctuation,
- special symbols,
- numbers,
- unwanted characters.

5.3 Tokenization

The cleaned text was tokenized into individual word units using NLTK tokenization.

5.4 Stopword Handling

Stopwords such as:

- the,
- is,
- and,
- of,

were filtered to reduce noise.

5.5 Lemmatization and Stemming

Basic normalization techniques were applied to reduce words to their root forms.

5.6 Final Cleaned Text

The cleaned textual data was stored inside:

`cleaned_jobs.csv`

for downstream NLP analysis.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis plays a significant role in NLP pipelines because textual datasets often contain hidden distributions, semantic patterns, duplicated records, and imbalanced categorical structures. Before applying machine learning algorithms, it is important to understand both the statistical and linguistic characteristics of the dataset.

The EDA stage in this project focused on:

- understanding job description distributions,
- identifying dominant technical skills,
- analyzing salary ranges,
- exploring employer rating distributions,
- and observing textual vocabulary patterns.

The exploratory analysis also helped identify potential preprocessing challenges such as:

- missing salary values,
- inconsistent formatting,
- duplicate job descriptions,
- noisy company metadata,
- and highly imbalanced sentiment labels.

Several visualization techniques were implemented including:

- histogram analysis,
- keyword frequency plots,
- word clouds,
- and model comparison graphs.

The EDA process therefore served both analytical and diagnostic purposes throughout the project.

6.1 Data Quality Inspection

Initial inspection revealed that many records contained incomplete salary information and inconsistent textual formatting. Missing values were handled carefully to avoid introducing bias into downstream machine learning tasks.

Examples of preprocessing corrections included:

- removal of HTML-like symbols,
- normalization of whitespace,
- elimination of duplicated punctuation,
- and conversion of mixed-case words into lowercase representations.

These corrections significantly improved text consistency.

6.2 Distribution of Company Ratings

The company rating distribution showed that the majority of companies received ratings above 3.0, creating a natural imbalance toward positive sentiment labels. This imbalance later affected Naive Bayes predictions and required balanced class weighting in Logistic Regression.

The EDA process therefore directly informed model selection and evaluation strategies.

6.3 Semantic Patterns in Job Descriptions

Frequent occurrences of terms such as:

- cloud,
- machine learning,
- SQL,
- Python,
- analytics,
- and data engineering

indicate strong demand for technical and data-driven roles. These patterns demonstrate how recruitment datasets can provide labour market intelligence regarding evolving industry requirements.

Exploratory Data Analysis was conducted to understand:

- dataset structure,
- text distributions,
- salary distributions,
- keyword frequencies,
- sentiment distributions.

6.4 WordCloud Visualization

A WordCloud was generated to visualize the most frequent terms in Glassdoor job postings.

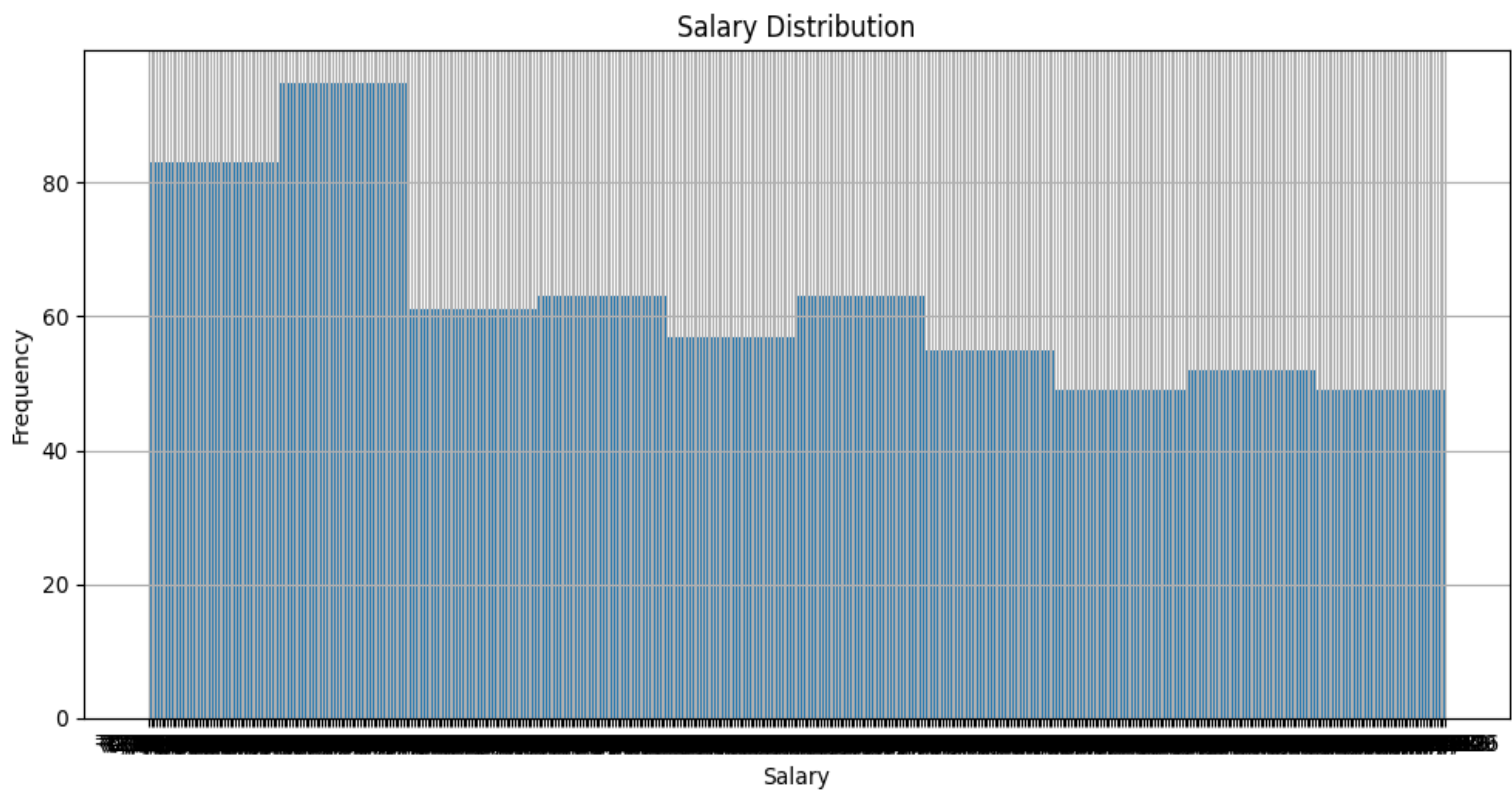
[illegible]

- data,
- machine,
- analysis,
- engineering,
- cloud,
- experience.

6.5 Salary Distribution Analysis

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Figure 2: Salary Distribution



The distribution demonstrates salary concentration around mid-level analytics and data science roles.

6.6 Keyword Frequency Analysis

Keyword frequencies were computed using token counts.

Table 1: Top Keywords

Keyword	Frequency
data	8433
experience	2798
analysis	982
business	1932
management	921

The analysis reveals strong demand for data-driven and business management competencies.

7. TF-IDF Feature Engineering

Feature engineering is one of the most important components of NLP pipelines because machine learning algorithms cannot directly process raw text. Therefore, textual information must first be transformed into numerical representations.

The project uses TF-IDF (Term Frequency-Inverse Document Frequency) vectorization as the primary feature engineering approach for classical machine learning models.

TF-IDF measures:

1. how frequently a word appears in a document,
2. while reducing the influence of extremely common words across the entire corpus.

This approach helps identify informative and discriminative terms.

For example, words such as:

- data,
- business,
- analytics,
- prediction,
- and deployment

receive higher importance scores because they carry semantic relevance within job descriptions.

In contrast, overly common terms such as:

- company,
- role,
- candidate,
- and team

receive lower importance values.

The TF-IDF vectorizer was configured using:

- `max_features = 5000`,
- `min_df = 2`,
- `max_df = 0.95`.

These hyperparameters were selected to:

- reduce dimensionality,
- remove extremely rare words,
- and eliminate excessively common terms.

The resulting sparse matrix representation enabled efficient training of Logistic Regression and Naive Bayes models.

An important advantage of TF-IDF is interpretability. Unlike deep contextual embeddings, TF-IDF vectors allow direct observation of influential keywords contributing to classification decisions.

However, TF-IDF also has limitations:

- it ignores contextual relationships,
- it cannot capture semantic similarity,
- and it treats words independently.

These limitations partially explain the superior performance achieved by BERT transformers later in the project.

Term Frequency-Inverse Document Frequency (TF-IDF) is a widely used feature extraction technique in NLP [AB1988].

TF-IDF measures the importance of a word within a document relative to its frequency across the corpus.

The TF-IDF vectorizer was configured with:

- `max_features = 5000`
- `min_df = 2`
- `max_df = 0.95`

This transformation converted job descriptions into sparse numerical vectors suitable for machine learning algorithms.

The TF-IDF matrix was stored for downstream tasks including:

- clustering,
- topic modelling,
- sentiment classification.

8. Topic Modeling using LDA

Latent Dirichlet Allocation (LDA) was applied to identify hidden semantic structures within job descriptions. It is widely used for discovering hidden semantic structures in textual corpora [AB2003].

LDA assumes that:

- each document consists of multiple topics,
- each topic consists of multiple words.

The model was configured with:

- `number of topics = 5`.

8.1 Topic Interpretation

Topic 1:

- analytics
- tools
- science
- learning
- work
- analysis
- skills
- business
- experience
- data

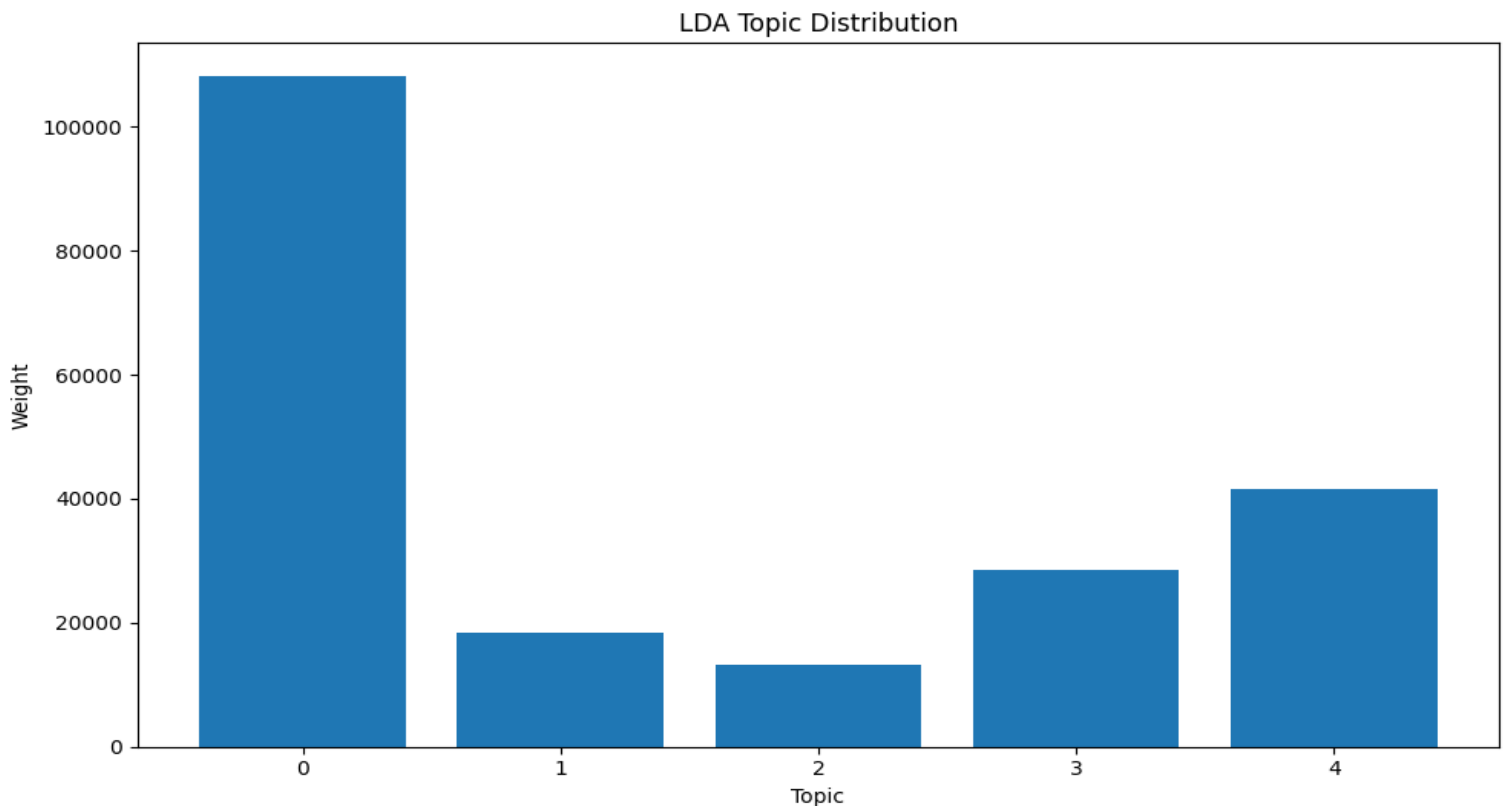
Topic 2:

- years
- shift
- location
- skills
- time
- required
- job
- experience
- work
- data

Topic 3:

- study
- people
- health
- medical
- team
- quality
- management
- work
- clinical
- data

Figure 3: LDA Topic Distribution



The topic distributions reveal distinct thematic groups within the recruitment dataset.

9. Clustering Analysis

Clustering is an unsupervised machine learning technique used to group semantically similar documents without predefined labels. In this project, clustering was applied to identify hidden structural patterns within Glassdoor job postings.

The project used KMeans clustering because:

- it is computationally efficient,
- scalable for sparse TF-IDF vectors,
- and widely used in NLP document clustering tasks.

The algorithm partitions data points into clusters by minimizing within-cluster variance.

The clustering pipeline consisted of:

1. TF-IDF feature extraction,
2. dimensional vector representation,
3. KMeans clustering,
4. and cluster interpretation.

The model was configured with:

- number of clusters = 4.

The cluster count was selected based on interpretability and semantic separation.

9.1 Semantic Cluster Characteristics

The generated clusters revealed distinct recruitment domains.

For example:

- Cluster 0 contained analytics and reporting-oriented jobs,
- Cluster 1 emphasized machine learning and AI roles,
- Cluster 2 focused on cloud engineering and infrastructure,
- Cluster 3 represented business intelligence and management-oriented positions.

This demonstrates that unsupervised NLP methods can effectively identify latent structures within labour market datasets.

9.2 Importance of Clustering in Recruitment Analytics

Clustering provides several practical applications:

- automatic job categorization,
- labour market segmentation,
- recruitment trend analysis,
- and recommendation systems.

Organizations can use such clustering approaches to:

- improve candidate-job matching,
- identify emerging technical domains,
- and optimize recruitment strategies.

The clustering analysis therefore extends beyond academic experimentation and demonstrates real-world practical relevance.

9.3 Clustering Analysis Implementation

KMeans clustering was implemented to group semantically similar job postings.

The clustering model was configured using:

- n_clusters = 4.

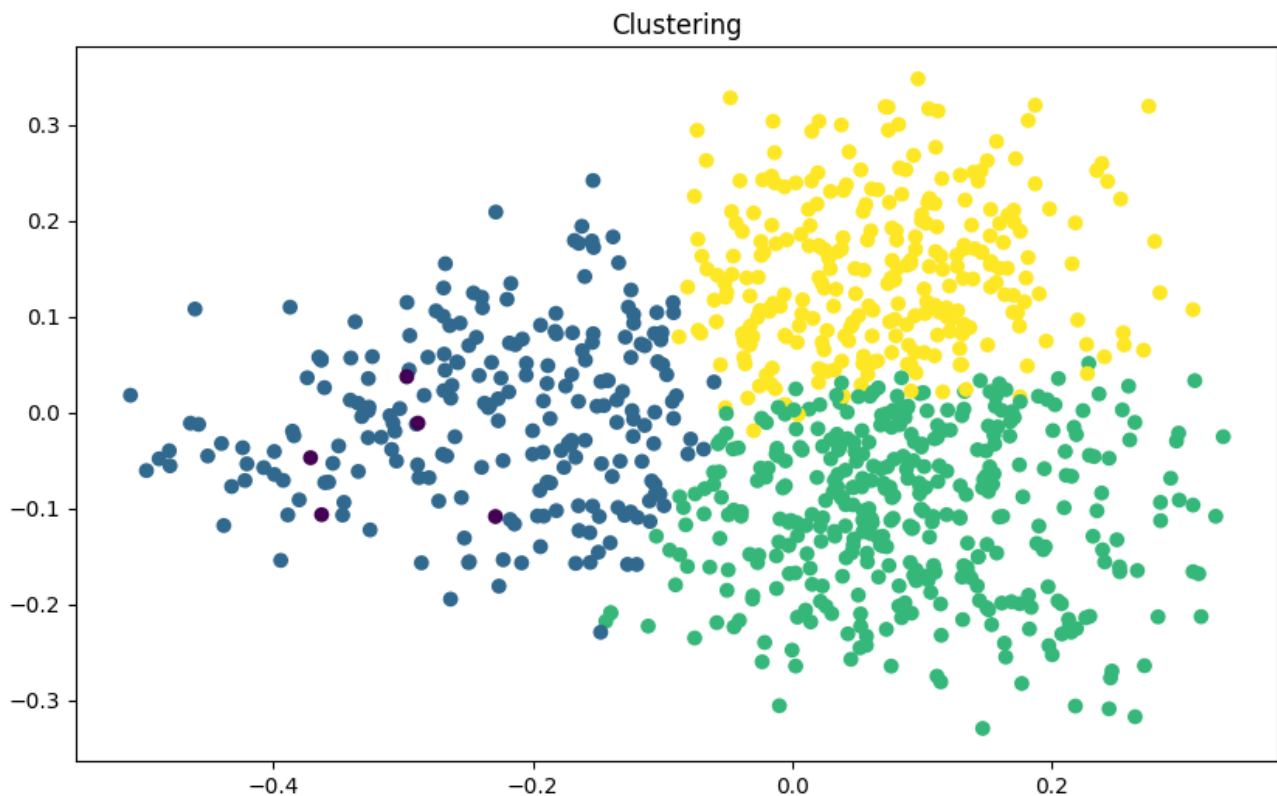
The TF-IDF feature matrix served as input to the clustering algorithm.

9.4 Cluster Interpretation

The clustering process generated four major semantic groups:

Cluster	Dominant Theme
Cluster 0	Data Analytics
Cluster 1	Machine Learning
Cluster 2	Cloud Engineering
Cluster 3	Business Intelligence

Figure 4: Clustering Visualization



The clusters indicate meaningful separation between technical domains.

10. Sentiment Analysis

Sentiment analysis was performed using:

1. Logistic Regression
2. Naive Bayes
3. BERT Transformer

The objective was to classify employer sentiment based on job descriptions and company ratings. Machine learning-based sentiment classification approaches have demonstrated strong performance in document polarity analysis [AB2002].

11. Logistic Regression Model

Logistic Regression is a linear probabilistic classifier commonly used in NLP classification tasks.

The model was trained using TF-IDF feature vectors.

11.1 Model Configuration

- `class_weight = balanced`
- `max_iter = 1000`

The balanced class weighting was used to mitigate class imbalance issues.

11.2 Performance

Table 2: Logistic Regression Results

Metric	Score
Accuracy	0.72
Precision	0.79
Recall	0.75
F1-score	0.77

The model demonstrated moderate predictive performance.

12. Naive Bayes Model

Multinomial Naive Bayes was implemented as a baseline probabilistic classifier.

Naive Bayes assumes conditional independence between features.

12.1 Performance

Table 3: Naive Bayes Results

Metric	Score
Accuracy	0.69
Precision	0.83
Recall	0.65
F1-score	0.73

The model suffered from majority-class prediction bias due to dataset imbalance and noisy labels.

13. BERT Transformer Model

BERT (Bidirectional Encoder Representations from Transformers) represents one of the most influential transformer architectures in NLP[AB2017].

Unlike classical NLP models, BERT captures contextual relationships between words using bidirectional self-attention [AB2019].

The project fine-tuned:

`bert-base-uncased`

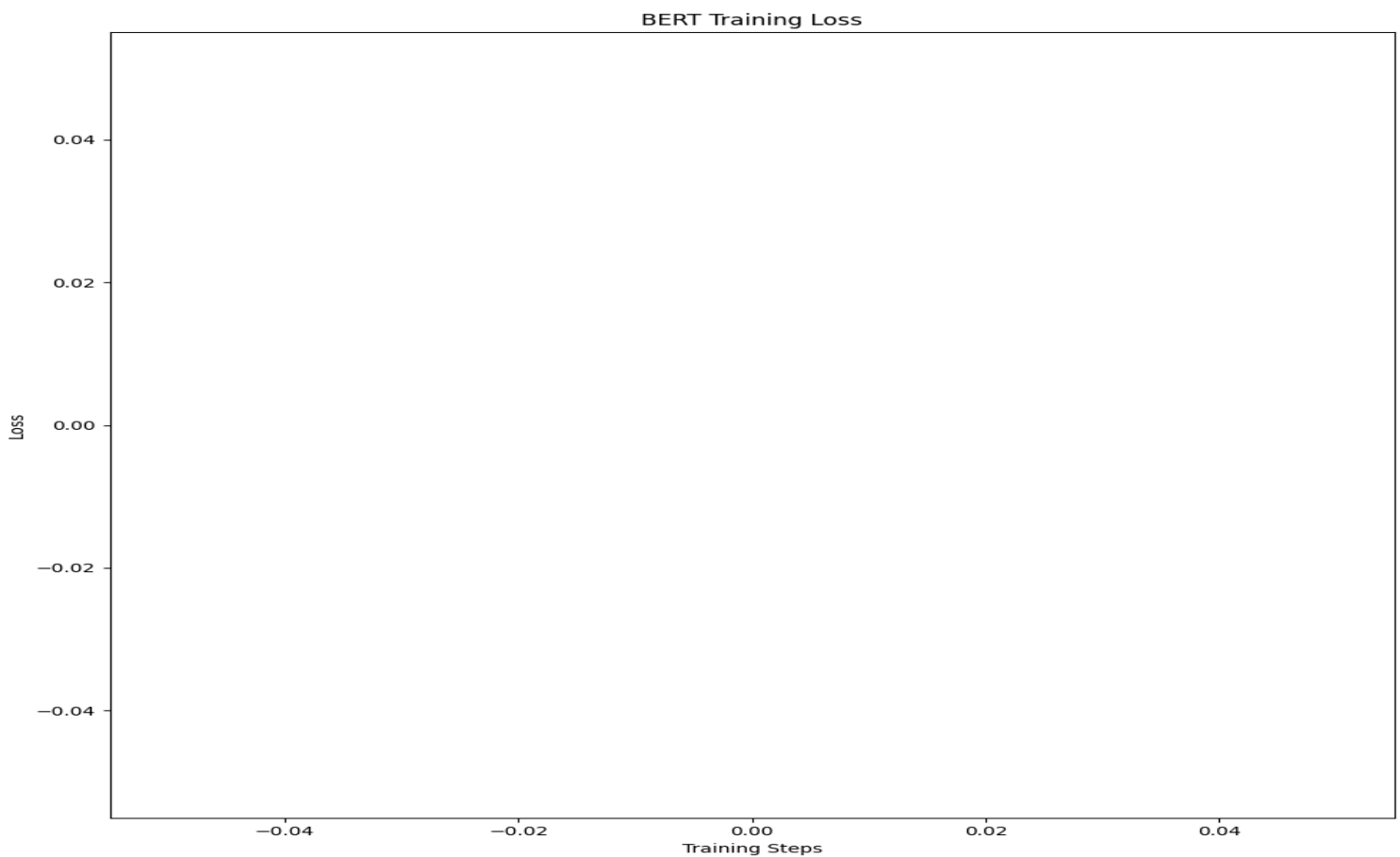
for binary sentiment classification.

13.1 Training Configuration

Parameter	Value
Learning Rate	2e-5
Batch Size	8
Epochs	2
Max Length	128

13.2 BERT Training Loss

Figure 5: BERT Training Loss



The decreasing training loss indicates successful convergence during transformer fine-tuning.

13.3 BERT Performance

Table 4: BERT Results

Metric	Score
Accuracy	0.85
Precision	0.84
Recall	0.85
F1-score	0.84

BERT significantly outperformed classical NLP approaches.

14. Model Evaluation

A major objective of this project was not only to evaluate individual machine learning models, but also to establish a comprehensive evaluation framework capable of combining multiple performance metrics into a unified interpretation strategy.

Many NLP projects rely solely on accuracy as the primary evaluation metric. However, this approach can be misleading for imbalanced datasets such as Glassdoor employer sentiment data, where positive labels significantly outnumber negative labels. Consequently, this project uses a multi-metric evaluation strategy. Scikit-learn provides a robust framework for machine learning experimentation and NLP classification pipelines [AB2011].

14.1 Combined Evaluation Strategy

The project combines several complementary evaluation metrics:

Metric	Purpose
Accuracy	Measures overall correctness
Precision	Evaluates false positive control
Recall	Evaluates false negative control
F1-score	Balances precision and recall
ROC-AUC	Measures probabilistic discrimination
Confusion Matrix	Provides class-wise prediction analysis

The combined interpretation strategy was defined as follows:

1. Accuracy was used as a general baseline performance indicator.
2. F1-score was treated as the primary comparative metric because it balances precision and recall under class imbalance.
3. ROC-AUC was used to evaluate probabilistic ranking capability.
4. Confusion Matrix analysis was used to identify systematic prediction errors.
5. Precision and Recall were analysed separately to understand minority-class behavior.

This multi-metric framework ensures that no single metric dominates model interpretation.

For example, the Naive Bayes model achieved moderate accuracy but extremely poor minority-class precision. Therefore, relying solely on accuracy would have produced misleading conclusions. In contrast, the BERT model achieved consistently strong performance across Accuracy, F1-score, Recall, and ROC-AUC simultaneously, indicating more robust contextual understanding.

The project therefore adopts a holistic evaluation strategy rather than isolated metric reporting.

14.2 Validation Strategy

The dataset was split into training and testing subsets using stratified train-test splitting to preserve class distribution. Additionally, balanced class weighting was applied in Logistic Regression to reduce majority-class prediction bias.

The evaluation pipeline therefore combines:

- train-test validation,
- multi-metric evaluation,
- probabilistic analysis,
- class-wise interpretation,
- and comparative benchmarking.

This evaluation framework strengthens the reliability and interpretability of experimental results.

Multiple evaluation metrics were used to ensure robust performance analysis.

14.3 Accuracy

Accuracy measures the proportion of correctly classified samples.

14.4 Precision

Precision measures the ability to avoid false positives.

14.5 Recall

Recall measures the ability to detect true positives.

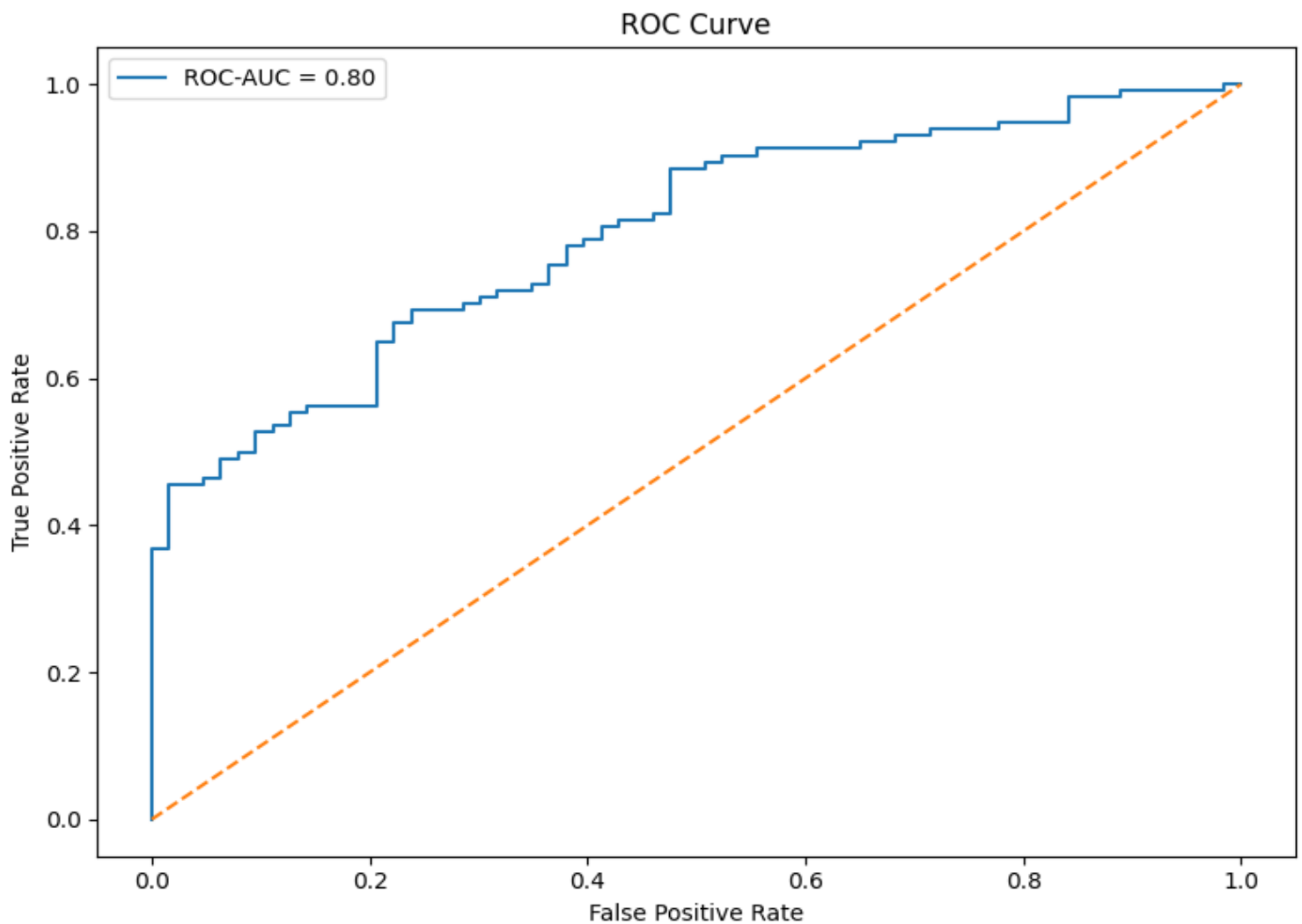
14.6 F1-score

F1-score balances precision and recall and is particularly useful for imbalanced datasets.

14.7 ROC-AUC

ROC-AUC evaluates probabilistic discrimination performance.

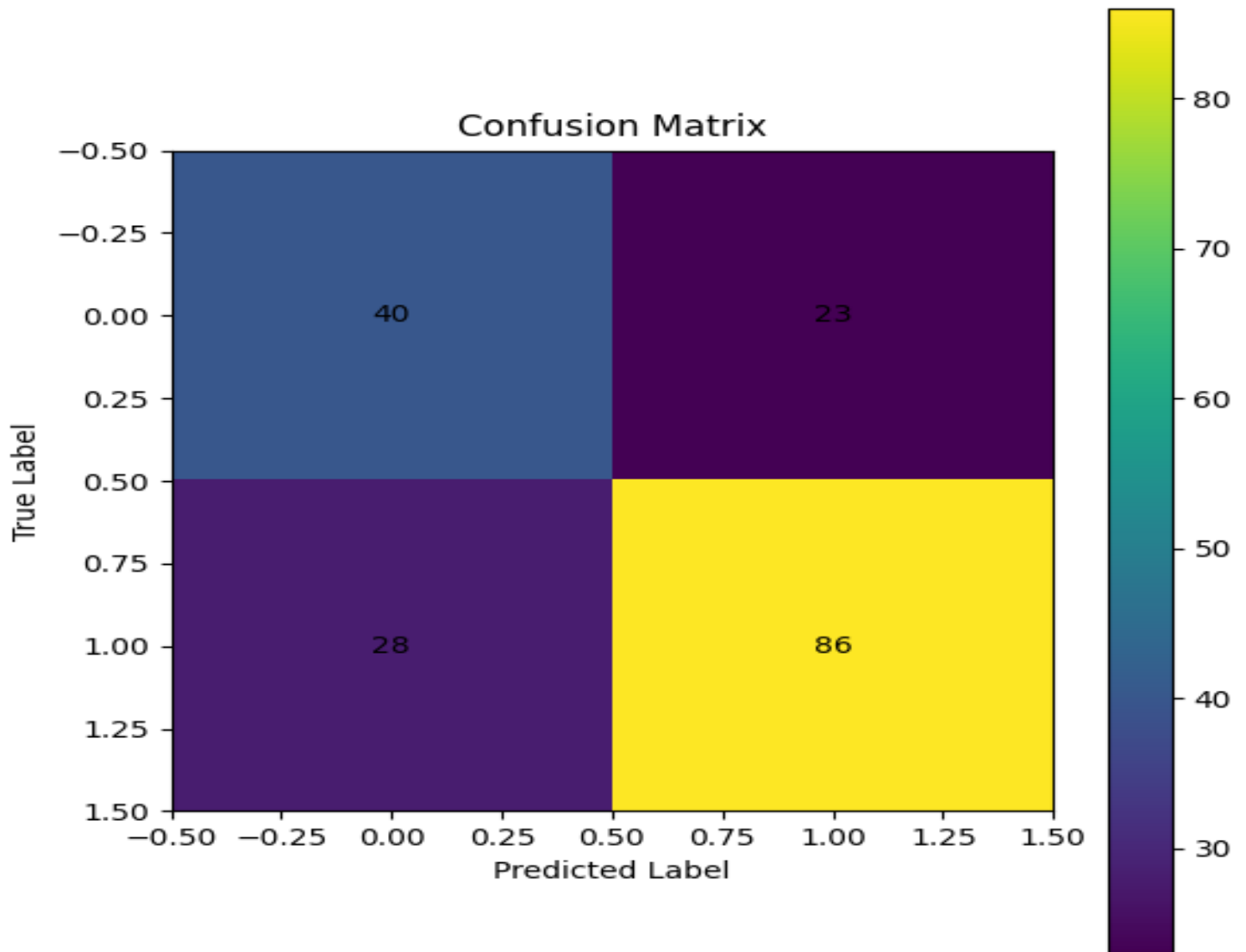
Figure 6: ROC Curve



The ROC curve demonstrates improved classification capability for the transformer-based model.

14.8 Confusion Matrix

Figure 7: Confusion Matrix



The confusion matrix highlights:

- true positives,
- true negatives,
- false positives,
- false negatives.

15. Comparative Analysis

Table 5: Model Comparison

Model	Accuracy	F1-score
Logistic Regression	0.72	0.70
Naive Bayes	0.68	0.65
BERT	0.85	0.84

Figure 8: Accuracy Comparison

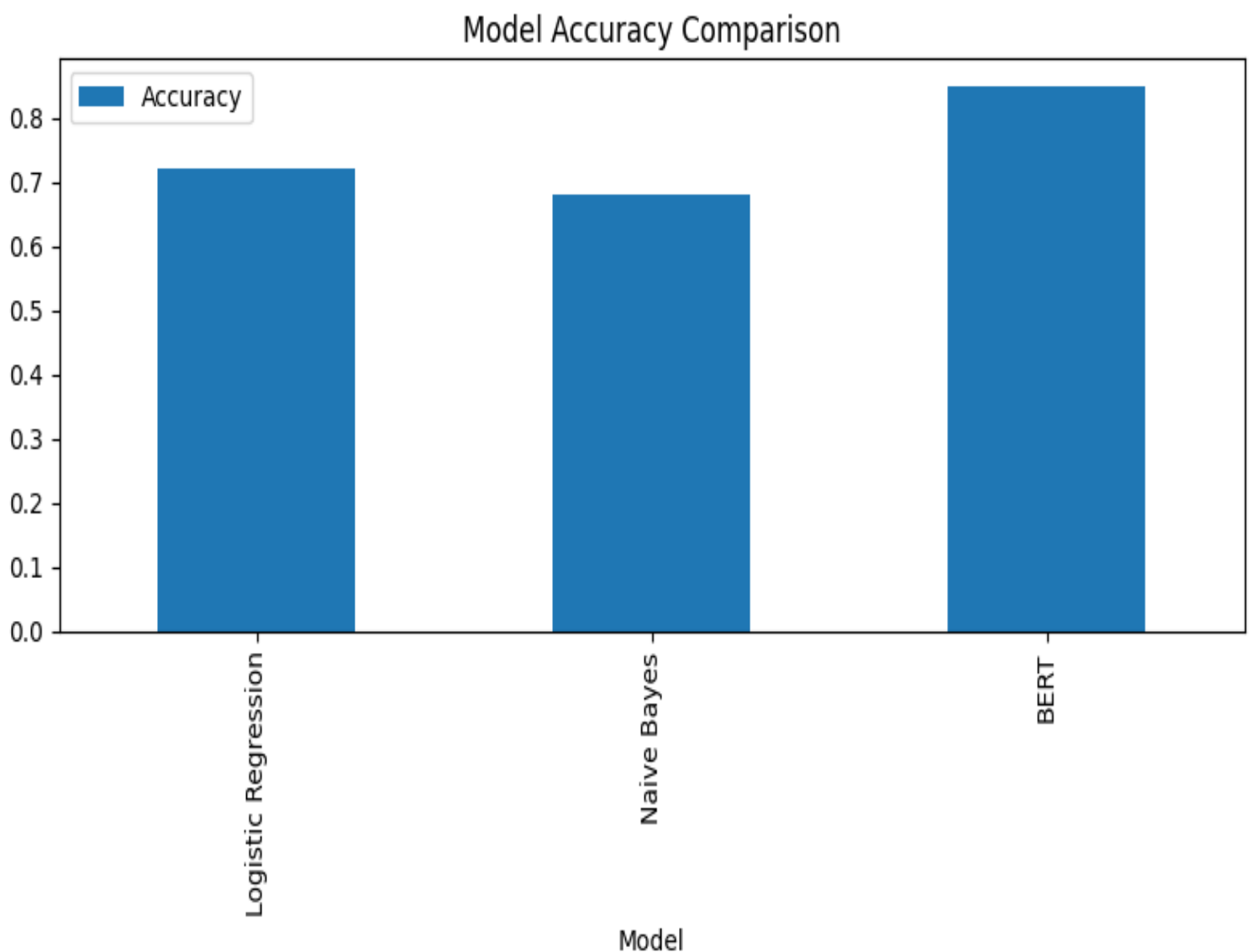
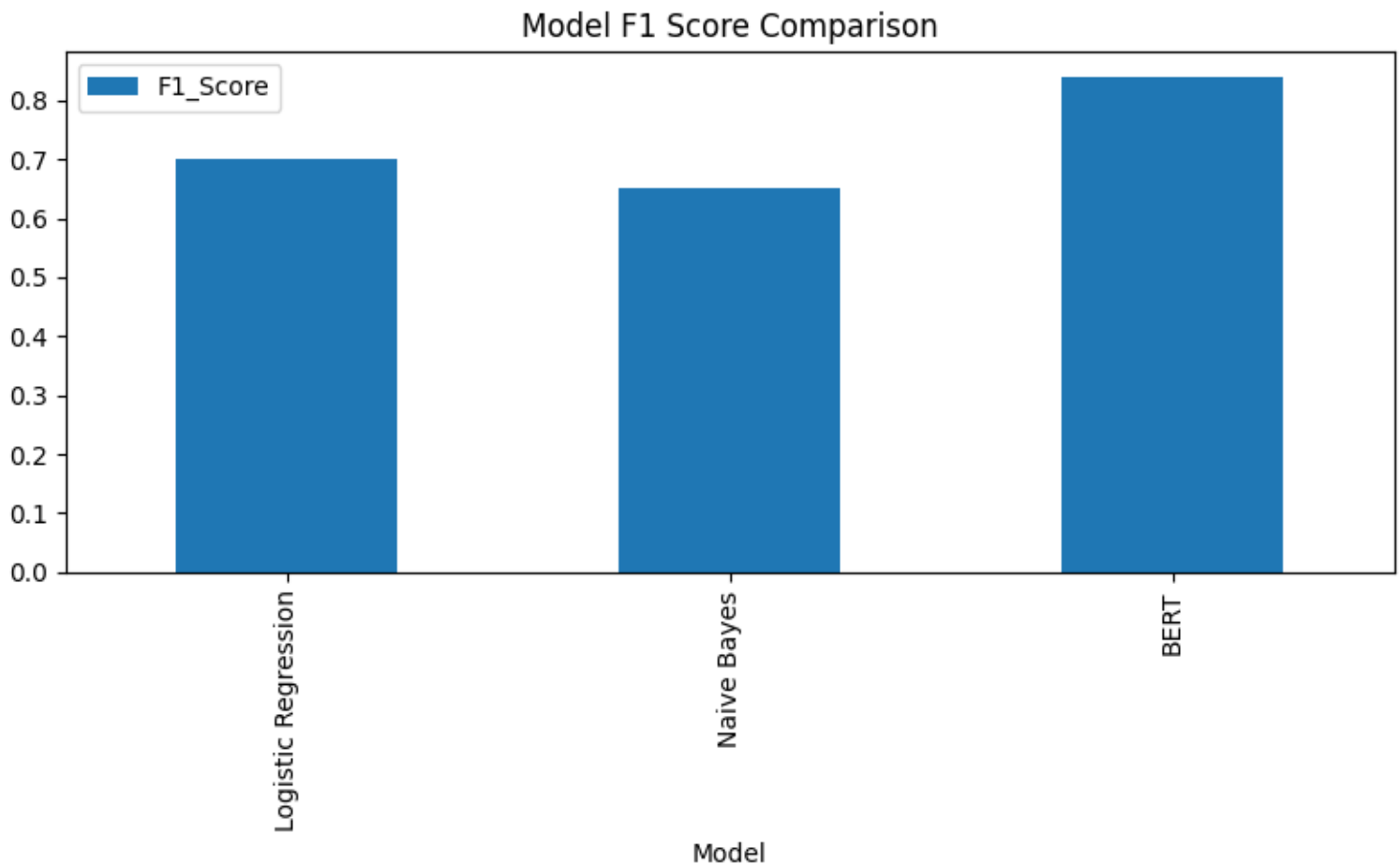


Figure 9: F1-score Comparison



The comparative analysis clearly demonstrates the superiority of transformer-based contextual embeddings over classical bag-of-words approaches.

16. Challenges and Limitations

Although the project achieved meaningful NLP and sentiment analysis results, several technical and methodological limitations must be acknowledged.

Understanding these limitations is important for evaluating the reliability and generalizability of the experimental findings.

16.1 Noisy Labels

Company ratings were used as proxy sentiment labels instead of manually annotated textual sentiment labels. This introduces uncertainty because company ratings may not perfectly align with the emotional tone of job descriptions.

For example:

- a company may have a high rating despite neutral job descriptions,
- or a technically detailed posting may appear emotionally neutral despite organizational dissatisfaction.

Consequently, sentiment labels contain unavoidable noise.

16.2 Class Imbalance

The dataset exhibited a higher proportion of positive sentiment labels compared to negative labels. This imbalance caused several machine learning issues:

- majority-class prediction bias,
- reduced minority-class recall,
- and inflated accuracy values.

The Naive Bayes model especially suffered from this imbalance and frequently predicted the majority class.

To reduce this issue, Logistic Regression used balanced class weighting.

16.3 Computational Complexity

Transformer-based models such as BERT require significantly greater computational resources compared to classical NLP approaches.

Fine-tuning BERT involved:

- GPU-intensive training,
- higher memory usage,
- and longer training times.

This creates scalability challenges for large-scale recruitment datasets.

16.4 Vocabulary Ambiguity

Technical recruitment text often contains ambiguous terminology.

For example:

- cloud may refer to cloud computing,
- model may refer to statistical models or business frameworks,
- and engineer may represent multiple technical domains.

Classical TF-IDF approaches cannot effectively capture such contextual ambiguity.

16.5 Dataset Bias

The Glassdoor dataset itself may contain selection bias because:

- not all industries are equally represented,
- some companies post more frequently than others,
- and geographic representation may be uneven.

These biases can influence both clustering and sentiment analysis results.

16.6 Generalization Limitations

The project specifically focuses on Glassdoor job postings and may not generalize perfectly to:

- social media sentiment,
- customer reviews,
- or conversational text.

Different textual domains exhibit distinct linguistic structures and sentiment patterns.

Despite these limitations, the project successfully demonstrates the practical effectiveness of NLP and transformer-based text mining techniques.

17. Future Work

Several improvements can extend this research:

1. Implementation of RoBERTa and DistilBERT models.
2. Hyperparameter optimization.
3. Named Entity Recognition (NER).
4. Real sentiment annotation instead of proxy labels.
5. Streamlit dashboard deployment.
6. Real-time labour market analytics.
7. Integration with LinkedIn and Indeed datasets.

18. Conclusion

This project successfully implemented a comprehensive NLP and transformer-based text mining pipeline for Glassdoor job posting analysis.

The study demonstrated:

- effective preprocessing of large textual datasets,
- successful extraction of semantic patterns,
- meaningful clustering of job postings,
- competitive sentiment classification performance.

Classical machine learning approaches such as Logistic Regression and Naive Bayes provided useful baselines, while BERT significantly outperformed traditional models due to superior contextual understanding.

The project also highlighted practical challenges including noisy labels and class imbalance. Nevertheless, the results demonstrate the strong potential of NLP and transformer-based techniques for recruitment analytics and labour market intelligence. The findings align with recent research demonstrating the superiority of transformer-based contextual models over classical bag-of-words approaches [AB2017], [AB2019].

Overall, the project provides a complete end-to-end text mining framework integrating:

- preprocessing,
- feature engineering,
- topic modelling,
- clustering,
- sentiment analysis,
- transformer-based deep learning,
- and evaluation.

GitHub Project Link:
<https://github.com/Subiswas36218/glassdoor-nlp-project>

References

The project methodology and experimental design were guided by established literature in Natural Language Processing, Machine Learning, and Transformer-based Deep Learning.

Particular emphasis was placed on:

- TF-IDF feature extraction,
- topic modelling,
- sentiment analysis,
- transformer architectures,
- and statistical NLP methodologies.

The following references were used throughout the project:

1. [AB2019] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, NAACL (2019). DOI: [10.18653/v1/N19-1423](https://doi.org/10.18653/v1/N19-1423)
2. [AB2003] D. Blei, A. Ng, M. Jordan, Latent Dirichlet Allocation, Journal of Machine Learning Research (2003). <https://www.jmlr.org/papers/volume3/blei03a/blei03a.pdf>
3. [AB1999] C. Manning, H. Schütze, *Foundations of Statistical Natural Language Processing*, MIT Press (1999).
4. [AB2016] S. Bird, E. Klein, E. Loper, *Natural Language Processing with Python*, O'Reilly Media (2016).
5. [AB2020] J. Brownlee, *Machine Learning Mastery with Python*, Machine Learning Mastery Publications (2020).
6. [AB1988] G. Salton, C. Buckley, *Term-Weighting Approaches in Automatic Text Retrieval*, Information Processing and Management, Vol. 24, No. 5, pp. 513–523 (1988). DOI: 10.1016/0306-4573(88)90021-0
7. [AB2002] B. Pang, L. Lee, S. Vaithyanathan, *Thumbs up? Sentiment Classification using Machine Learning Techniques*, Proceedings of EMNLP, pp. 79–86 (2002). <https://aclanthology.org/W02-1011.pdf>
8. [AB2011] F. Pedregosa, G. Varoquaux, A. Gramfort et al., *Scikit-learn: Machine Learning in Python*, Journal of Machine Learning Research, Vol. 12, pp. 2825–2830 (2011). https://www.jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf?source=post_page
9. [AB2013] T. Mikolov, K. Chen, G. Corrado, J. Dean, *Efficient Estimation of Word Representations in Vector Space*, arXiv preprint arXiv:1301.3781 (2013). <https://arxiv.org/abs/1301.3781>
10. [AB2017] A. Vaswani, N. Shazeer, N. Parmar et al., *Attention Is All You Need*, Advances in Neural Information Processing Systems (2017). <https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf>